

PHYSICS

PAPER 1 (THEORY)

Maximum Marks: 70**Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during reading time.
3. This question paper has 9 printed pages.
4. It has twenty questions in this paper. Answer all questions.
5. There are four sections in the paper: A, B, C and D. Internal choices have been provided in two questions each in Sections B, C and D.
6. Section A consists of one question having fourteen subparts of one mark each.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B consists of seven questions of two marks each.
9. Section C consists of nine questions of three marks each.
10. Section D consists of three questions of five marks each.
11. The intended marks for questions are given in brackets [].
12. A list of useful constants and relations is given after these instructions.
13. A simple scientific calculator without a programmable memory may be used for calculations.

USEFUL CONSTANTS AND RELATIONS

- | | | | | |
|-----|-----------------------------------|--------------------|---|---|
| 1. | Charge of an electron (magnitude) | e | = | $1.6 \times 10^{-19} \text{ C}$ |
| 2. | Planck's constant | h | = | $6.6 \times 10^{-34} \text{ J s}$ |
| 3. | Speed of light in vacuum | c | = | $3 \times 10^8 \text{ m s}^{-1}$ |
| 4. | Permittivity of vacuum | ϵ_0 | = | $8.85 \times 10^{-12} \text{ F m}^{-1}$ |
| 5. | | $1/4\pi\epsilon_0$ | = | $9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ |
| 6. | Permeability of vacuum | μ_0 | = | $4\pi \times 10^{-7} \text{ H m}^{-1}$ |
| 7. | One electron volt | 1 eV | = | $1.6 \times 10^{-19} \text{ J}$ |
| 8. | Atomic mass unit | 1 u | = | $931.5 \text{ MeV}/c^2 = 1.66 \times 10^{-27} \text{ kg}$ |
| 9. | Mass of a hydrogen atom | m_{H} | = | 1.007825 u |
| 10. | Mass of a neutron | m_{n} | = | 1.008665 u |

Instruction to Supervising Examiner: Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

SECTION A – 14 MARKS

Question 1

(A) In questions (i) to (vii), choose the correct alternative (a), (b), (c) or (d) for each of the questions given below.

- (i) Figure 1 shows a closed surface S . Charges $+3q$ and $-q$ are inside it and a charge $+2q$ is outside it. The net electric flux through S is: (*Apply*) [1]

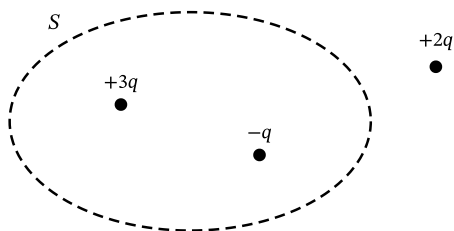


Figure 1

- (a) $\frac{4q}{\epsilon_0}$
(b) $\frac{2q}{\epsilon_0}$
(c) $\frac{5q}{\epsilon_0}$
(d) zero
- (ii) Select the odd one out from the following materials. (*Analyse*) [1]
(a) Bismuth
(b) Copper
(c) Aluminium
(d) Water
- (iii) A circular coil of N turns and radius R carries a current I and produces a magnetic field B at its centre. The same wire is unwound and rewound into a coil of radius $R/2$, carrying the same current. The field at the centre now is: (*Apply*) [1]
(a) $B/2$
(b) B
(c) $2B$
(d) $4B$

- (iv) Match the entries in Column I with those in Column II.

[1]

Column I	Column II
1. Davisson–Germer experiment	A. Wave nature of electrons
2. Photoelectric effect	B. Particle nature of light
3. Geiger–Marsden (α -scattering) experiment	C. Existence of a small, massive nucleus
4. Pair production	D. Conversion of energy into mass

(Recall)

- (a) 1-C, 2-B, 3-D, 4-A
- (b) 1-A, 2-B, 3-C, 4-D
- (c) 1-A, 2-C, 3-B, 4-D
- (d) 1-D, 2-B, 3-C, 4-A

- (v) Identify the correct statements about an n-type semiconductor.

[1]

- (P) Electrons are the majority charge carriers.
- (Q) It carries a net negative charge.
- (R) It is made by doping silicon with a pentavalent impurity such as arsenic.
- (S) Holes are the majority charge carriers.

(Understand)

- (a) Both (P) and (Q)
- (b) Both (P) and (R)
- (c) Both (Q) and (S)
- (d) Both (R) and (S)

- (vi) An object is placed 15 cm in front of a concave mirror of focal length 10 cm. The image formed is: (*Apply*)

[1]

- (a) virtual, erect and 30 cm behind the mirror
- (b) real, inverted, magnified and 30 cm in front of the mirror
- (c) real, inverted, diminished and 6 cm in front of the mirror
- (d) real, inverted, of the same size and 20 cm in front of the mirror

- (vii) Given below are two statements marked, Assertion and Reason. Read the two statements and choose the correct option.

[1]

Assertion: If the whole apparatus of Young's double-slit experiment is immersed in water, the fringe width decreases.

Reason: The wavelength of light in water is less than its wavelength in air.

(Analyse)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Assertion is false and Reason is true.

- (B) Answer the following questions briefly:
- (i) How much work is done in moving a charge of $5\ \mu\text{C}$ through 10 cm along the equatorial line of an electric dipole? (*Understand*) [1]
 - (ii) What is meant by *motional emf*? (*Recall*) [1]
 - (iii) State one advantage of a reflecting telescope over a refracting telescope. (*Recall*) [1]
 - (iv) An electron and a proton have the same de Broglie wavelength. Which of the two has the greater kinetic energy? (*Analyse*) [1]
 - (v) What is the minimum energy of a γ -ray photon that can produce an electron–positron pair? (*Recall*) [1]
 - (vi) Why is a photodiode operated in reverse bias? (*Understand*) [1]
 - (vii) Fill in the blank with an appropriate term from the bracket.
The phase difference between any two points on the same wavefront is _____ (zero, $\pi/2$, π , 2π per wavelength). (*Recall*) [1]

SECTION B – 14 MARKS

Question 2

- (i) Three capacitors of capacitance $2\ \mu\text{F}$, $3\ \mu\text{F}$ and $6\ \mu\text{F}$ are connected in series across a 12 V battery. Calculate the equivalent capacitance and the potential difference across the $2\ \mu\text{F}$ capacitor. (*Apply*) [2]

OR

- (ii) An infinitely long, thin straight wire carries a uniform linear charge density λ . The electric field at a distance of 2 cm from it is $9 \times 10^4\ \text{N C}^{-1}$. Calculate λ . (*Apply*) [2]

Question 3

- Draw a labelled circuit diagram of the arrangement used in the Physics laboratory to determine the internal resistance of a cell using a potentiometer. Write the formula used. (*Understand*) [2]

Question 4

- (i) A copper wire of cross-sectional area $1.0\ \text{mm}^2$ carries a current of 1.6 A. The number density of free electrons in copper is $8.0 \times 10^{28}\ \text{m}^{-3}$. Calculate the drift speed of the electrons. (*Apply*) [2]

OR

- (ii) Two bulbs rated 100 W, 220 V and 60 W, 220 V are connected in series across a 220 V supply. Which bulb glows brighter? Justify your answer. (*Analyse*) [2]

Question 5

- A galvanometer of resistance $50\ \Omega$ gives full-scale deflection for a current of 5 mA. How can it be converted into a voltmeter of range 0–10 V? Calculate the value of the resistance required. (*Apply*) [2]

Question 6

- (i) Name the electromagnetic radiation used in water purifiers to kill germs. State one source of it. (*Recall*) [1]

- (ii) Name the electromagnetic radiation used for taking photographs in haze and fog. Why is it suitable? (*Recall*) [1]

Question 7

- (i) What is meant by the *depletion region* of a p-n junction? (*Recall*) [1]
(ii) How is the potential barrier across the junction formed? (*Understand*) [1]

Question 8

Monochromatic light of a frequency above the threshold frequency falls on a photosensitive surface. How will (i) the photoelectric current and (ii) the maximum kinetic energy of the photoelectrons change if the intensity of light is doubled? Give a reason in each case. (*Understand*) [2]

SECTION C – 27 MARKS

Question 9

- (i) Obtain an expression for the electric field intensity at a point on the axial line of an electric dipole. Hence write it for a short dipole. (*Understand*) [3]

OR

- (ii) An electric dipole of moment p is held in a uniform electric field E with its axis at an angle θ to the field. Show the forces acting on the charges and hence prove that the torque on it is $\tau = pE \sin \theta$. (*Understand*) [3]

Question 10

- (i) In the network shown in Figure 2, a cell of emf 6 V and internal resistance $1\ \Omega$ and a cell of emf 4 V and internal resistance $2\ \Omega$ are connected in parallel across a $4\ \Omega$ resistor. Using Kirchhoff's laws, calculate the current in each branch and the potential difference between A and B. (*Apply*) [3]

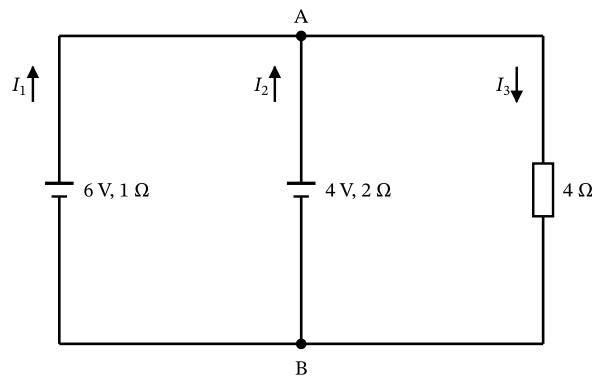


Figure 2

OR

- (ii) In the metre bridge shown in Figure 3, an unknown resistance X is in the left gap and a $10\ \Omega$ and a $15\ \Omega$ resistor are connected in parallel in the right gap. The null point is at 40 cm from end A. (a) Calculate X . (b) Where will the null point be if the $15\ \Omega$ resistor is removed? (*Apply*) [3]

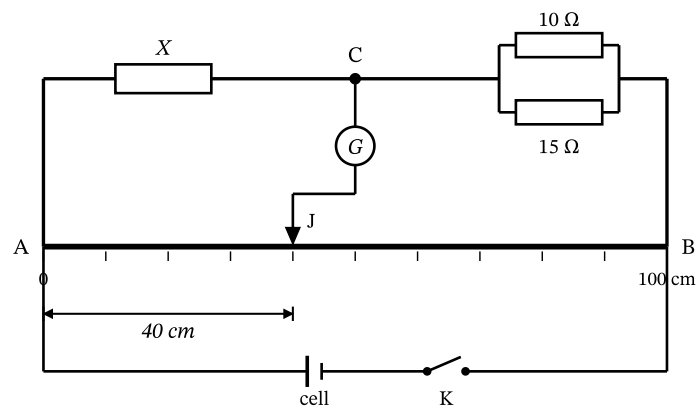


Figure 3

Question 11

Using the Biot–Savart law, obtain an expression for the magnetic field at the centre of a circular coil of radius R having N turns and carrying a current I . (*Understand*) [3]

Question 12

- (i) Two long straight parallel wires 10 cm apart carry currents of 5 A and 10 A in the same direction. Calculate the force per unit length on each wire and state whether it is attractive or repulsive. (*Apply*) [2]
- (ii) Why is soft iron preferred to steel for the core of a transformer? (*Recall*) [1]

Question 13

A physics teacher asked the following questions about lenses. State whether each student's answer is correct or incorrect. Correct an incorrect response with a brief explanation.

- (i) **Teacher:** A convex lens forms a sharp image of a candle on a screen. What happens to the image if the lower half of the lens is covered with black paper? [1]
Student: The lower half of the image disappears. (*Evaluate*)
- (ii) **Teacher:** What happens to the focal length of a convex lens if red light is replaced by violet light? [1]
Student: It increases. (*Evaluate*)
- (iii) **Teacher:** Two thin lenses of power +5 D and -3 D are placed in contact. What is the nature and focal length of the combination? [1]
Student: Converging, of focal length 50 cm. (*Evaluate*)

Question 14

- (i) A prism of angle 60° is made of glass of refractive index $\sqrt{2}$. Calculate the angle of minimum deviation and the angle of incidence at which it occurs. (*Apply*) [2]
- (ii) When white light passes through the prism, which colour is deviated the most? Why? (*Understand*) [1]

Question 15

Using Huygens' principle, draw a labelled diagram showing the reflection of a plane wavefront at a plane surface, and hence prove the laws of reflection. (*Understand*) [3]

Question 16

Figure 4 shows the variation of stopping potential V_s with the frequency ν of incident radiation for two metals A and B. Study it and answer the questions.

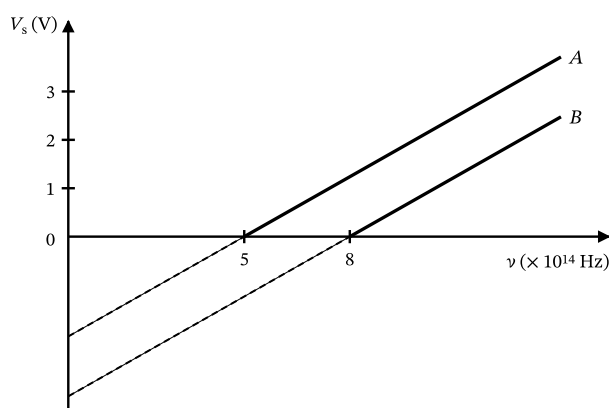


Figure 4

- (i) Which metal has the greater work function? Give a reason. (*Analyse*) [1]
- (ii) Why are the two graphs parallel to each other? (*Understand*) [1]
- (iii) Calculate the work function of metal A in eV. (*Apply*) [1]

Question 17

Draw a labelled circuit diagram of a full-wave rectifier using two junction diodes. Show the input and output waveforms and explain briefly how it works. (*Understand*) [3]

SECTION D – 15 MARKS

Question 18

- (i) (a) A series LCR circuit has $R = 40\ \Omega$, $L = 80\text{ mH}$ and $C = 20\ \mu\text{F}$. It is connected to an ac source of emf $e = 250\sqrt{2} \sin(1000t)\text{ V}$. Calculate (1) the impedance of the circuit, (2) the rms current and (3) the phase angle between the current and the source voltage, stating which leads. (Apply) [3]
- (b) At what frequency of the source will the current in this circuit be maximum? What is the power factor then? (Apply) [2]

OR

- (ii) (a) The current in a coil falls steadily from 5 A to zero in 20 ms and an emf of 250 V is induced in it. Calculate the self-inductance of the coil. (Apply) [2]
- (b) Why is the core of a transformer laminated? (Understand) [1]
- (c) An ideal step-up transformer raises 220 V to 11 000 V. Its primary has 100 turns. Calculate the number of turns in the secondary, and the primary current when the secondary supplies 0.2 A. (Apply) [2]

Question 19

- (i) (a) State Bohr's postulates for the circular motion of the electron in a hydrogen atom and the quantisation of its angular momentum. Using them, obtain an expression for the radius of the n th orbit. (Apply) [3]
- (b) Prove that $1\text{ u} = 931.5\text{ MeV}$, where the terms have their usual meaning. (Apply) [2]

OR

- (ii) (a) Taking the radius of a nucleus as $R = R_0 A^{1/3}$, show that the density of nuclear matter is independent of the mass number A . (Apply) [2]
- (b) Define *isotopes* and *isobars*, giving one example of each. (Recall) [1]
- (c) Calculate the binding energy per nucleon of ${}^4_2\text{He}$. Mass of ${}^4_2\text{He}$ atom = 4.002603 u. (Apply) [2]

Question 20

Read the passage and answer the questions that follow.

A public aquarium displays a small fish in a spherical glass bowl of radius 30 cm filled with water (refractive index $4/3$); the glass is thin enough to be ignored. A visitor looks at the fish along a diameter, from outside, when the fish is 10 cm inside the wall nearest to her (Figure 5). The aquarium's underwater camera uses a glass lens ($n = 1.5$) of focal length 5 cm in air, and its display tanks are lit by light carried through optical fibres.

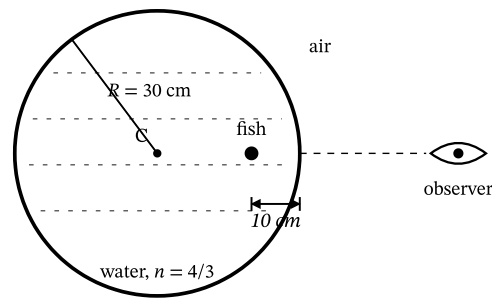


Figure 5

- (i) Where does the fish appear to the visitor? (*Apply*) [2]
- (ii) What is the focal length of the camera lens when it is used under water? (*Apply*) [1]
- (iii) Calculate the critical angle for light going from the water to air. (*Apply*) [1]
- (iv) State the principle on which the optical fibres work, and the condition on the refractive index of the core. (*Recall*) [1]

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